Unraveling Telangana's Path to Inclusive Development

Final Report for: The BDM Capstone Project

Submitted by: Jishnu S G

Roll Number: 21f1006877

Institution: Indian Institute of Technology, Madras

About the Organization & Data

- Telangana is a south-central state in Indian subcontinent with rich history, diverse culture and rapid growth.
- States Vision : A leader in data-driven governance, committed to balancing industrial growth and agrarian sustainability.
- Open Data Telangana : An initiative from government to enhance transparency and data accessibility .
- This analysis is based on data scraped from this public portal.
- Key Datasets : We analyzed TS-iPASS, Vehicle Registrations, Land Registrations, and Weather data covering the period from 2019 to 2024.

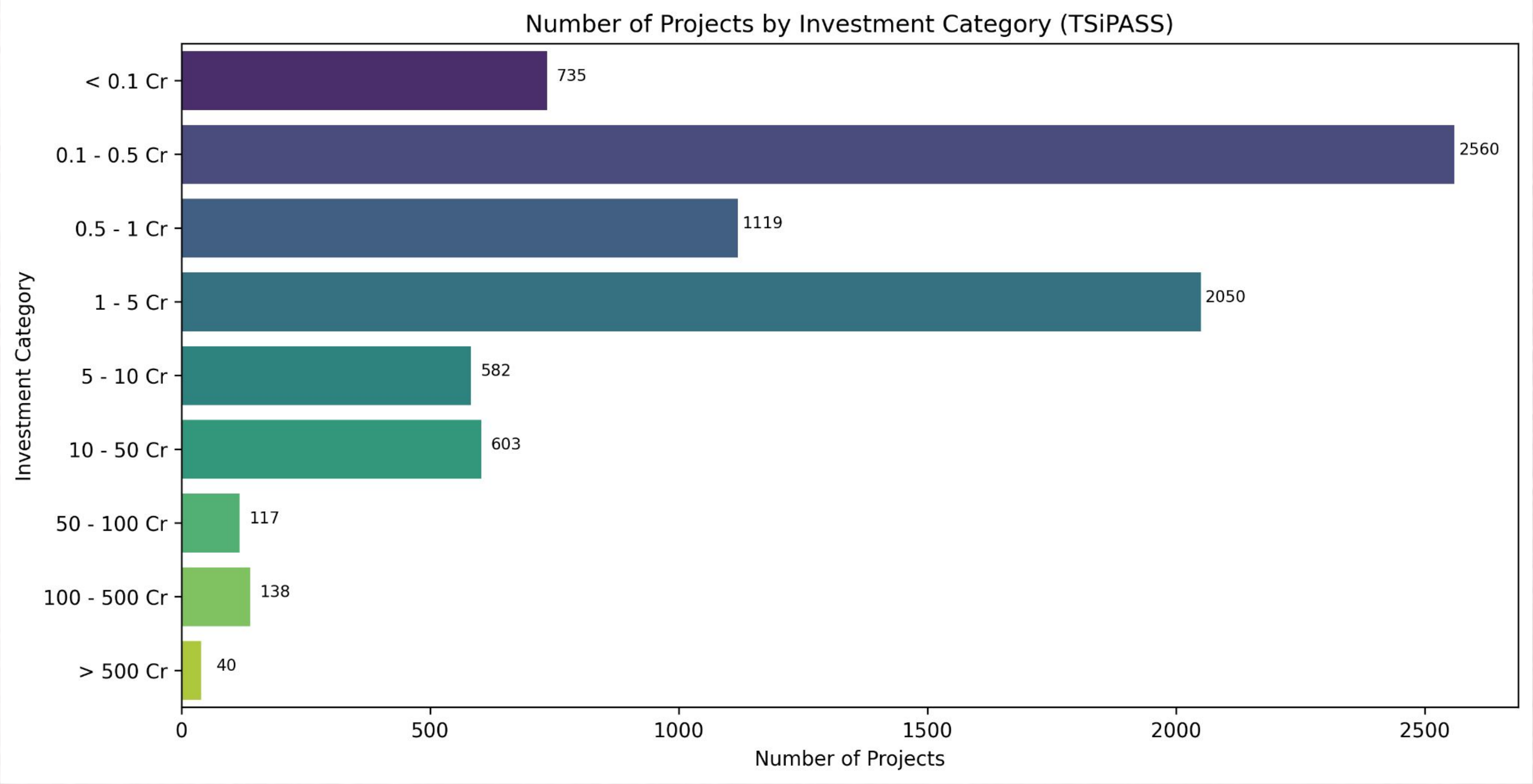
Key Problems & Objectives

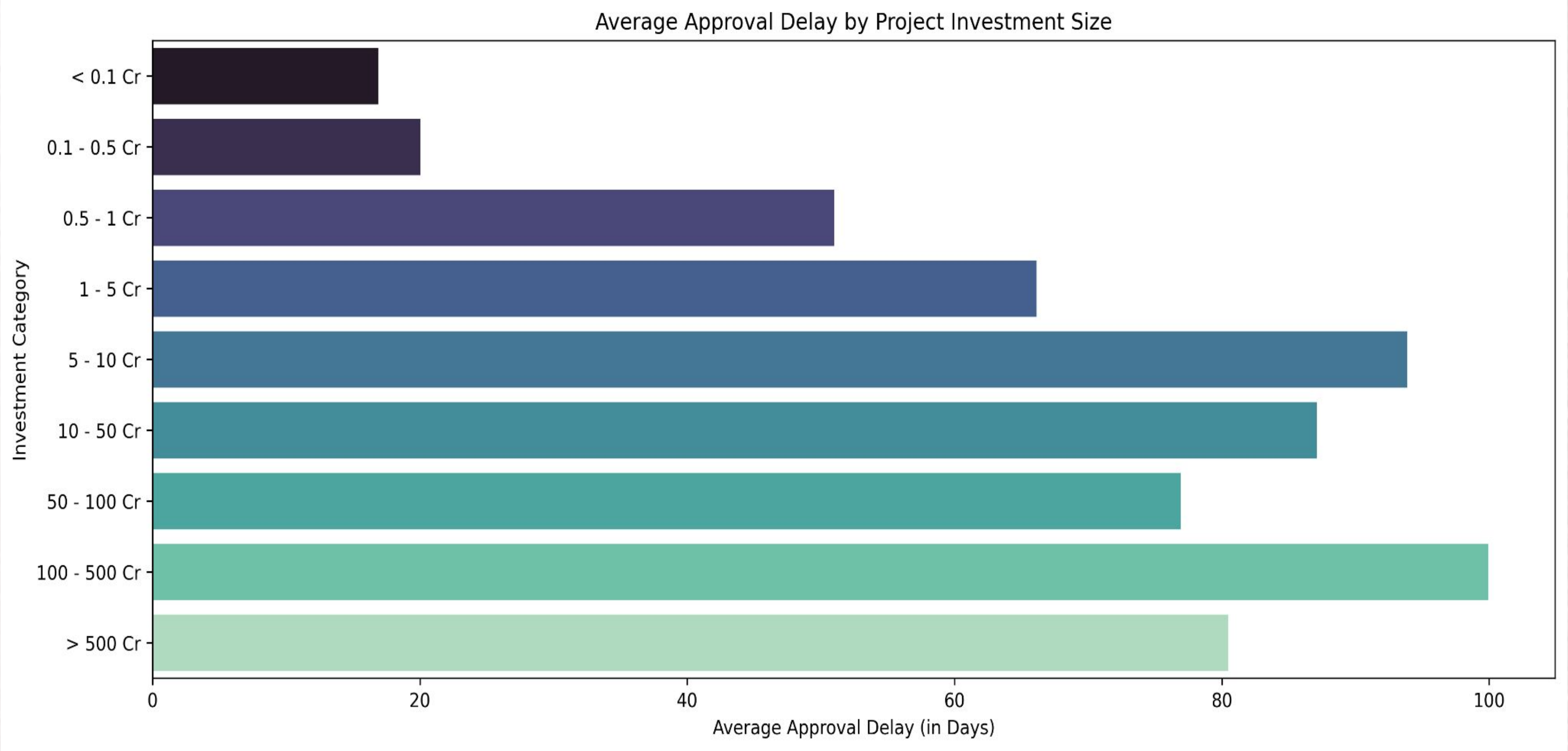
- **Key Problems**
    - **Problem 1** : Small investments face systemic approval delays : Smaller projects may experience disproportionately longer approval timelines compared to large-scale proposals, discouraging local entrepreneurship.
    - **Problem 2** : e-Stamp adoption efficiency over document registration : Identify if digital transitions in registrations improved revenue collection efficiency for scalable governance and promoting digital inclusion.
    - **Problem 3** : Rainfall-mechanization mismatch inflates farm costs : Uneven rainfall forces over/under investments in agri-vehicles without promising irrigation infrastructure might lead to unnecessary capital expenditure and subsidies.
- **Objective**
    - To provide data-backed recommendations that ensure inclusive growth and climate resilience.
    - To bridge the urban-rural divide by aligning government policies with real-world data.

Data and Methodology

- **Data Sourcing**
  - We used a web-scraping script to collect data from the data.telangana.gov.in portal in tabular format.
- **Data Cleaning**
  - The raw data was meticulously cleaned and unified to handle inconsistencies in naming, dates, and categories.
  - Python and pandas module for the data cleaning and filtering data from 2019 to 2024 only.
- **Analysis**
  - We mainly used time-series, statistical and comparative analytics to generate district-level insights from the data.
- **Visualizations**
  - Key findings are presented using plots from Matplotlib, Seaborn, and Plotly.

Key Findings: Problem 1

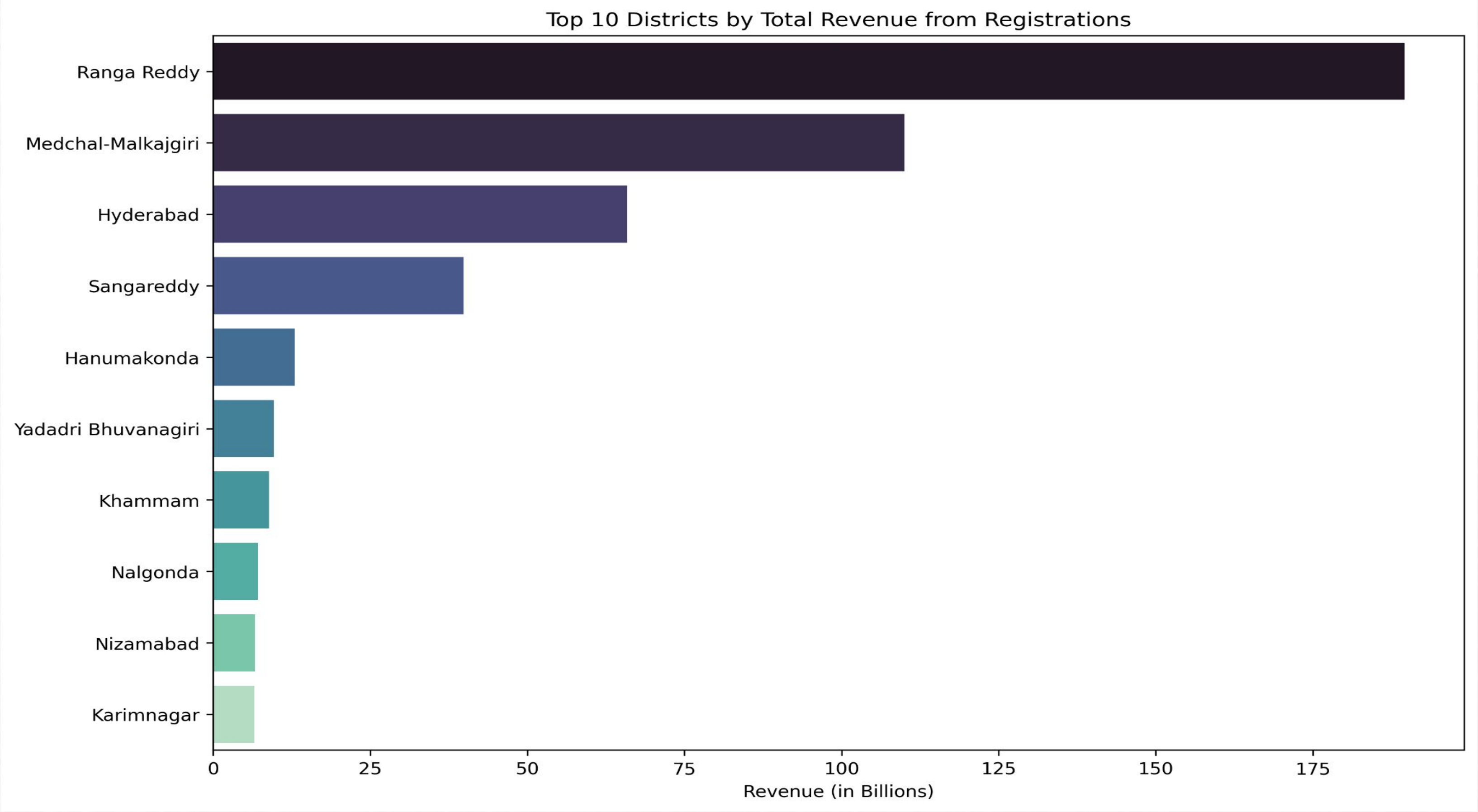


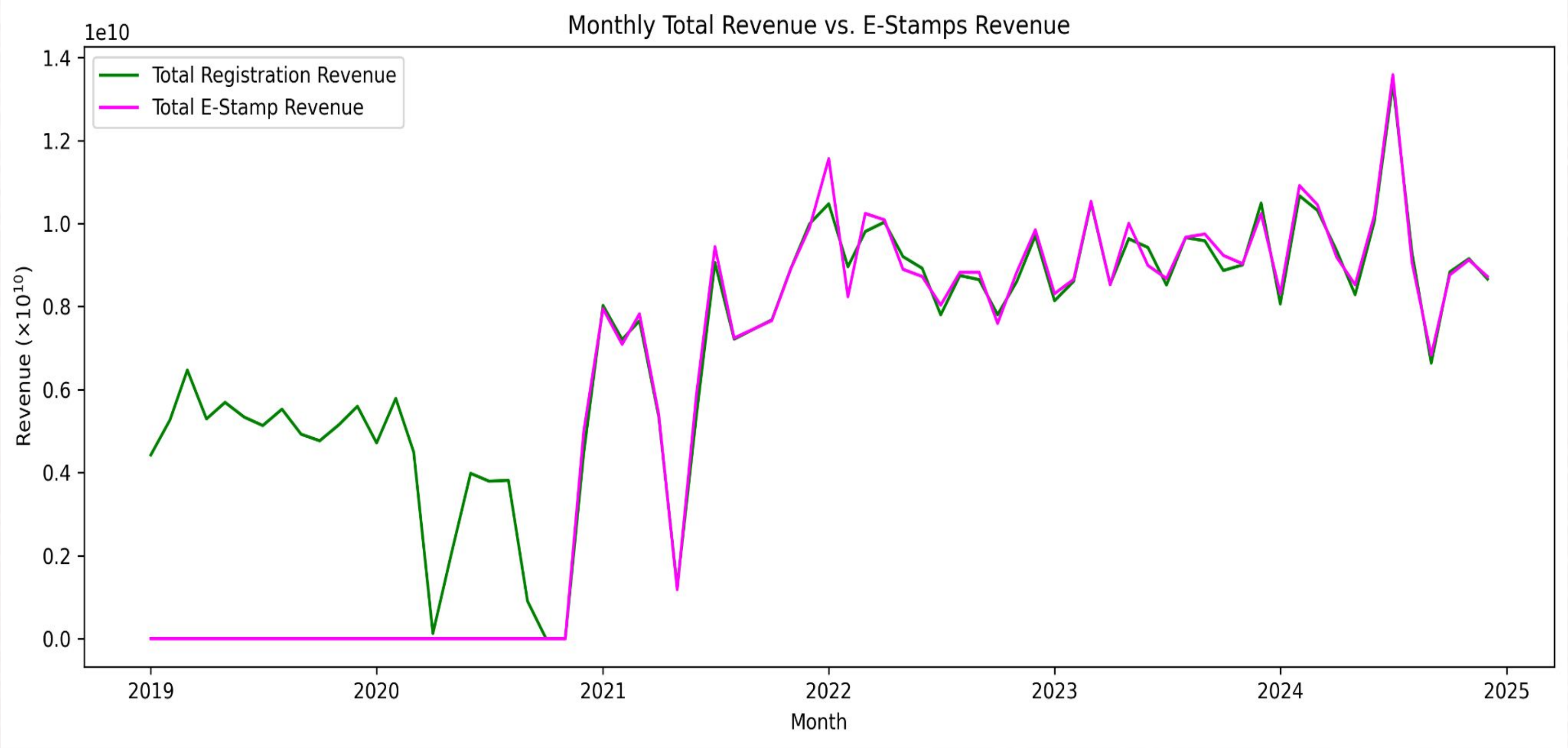


Small businesses face approval delays, which may limit entrepreneurship.

- **Initial Hypothesis**
  - The portrayal of states growth was shown as the large scale investments.
  - This led us to assume MSMEs faces systemic delays.
- **Data-Driven Finding** :
  - Most projects are of investments under 5 crores.
  - Job creation doesn’t directly scales with investment size.
  - Direct analysis of approval times revealed the opposite.
  - The longest approval delays are for the largest projects, not small businesses.
  - Approval timeline is correlated with type of sector.

Key Findings: Problem 2

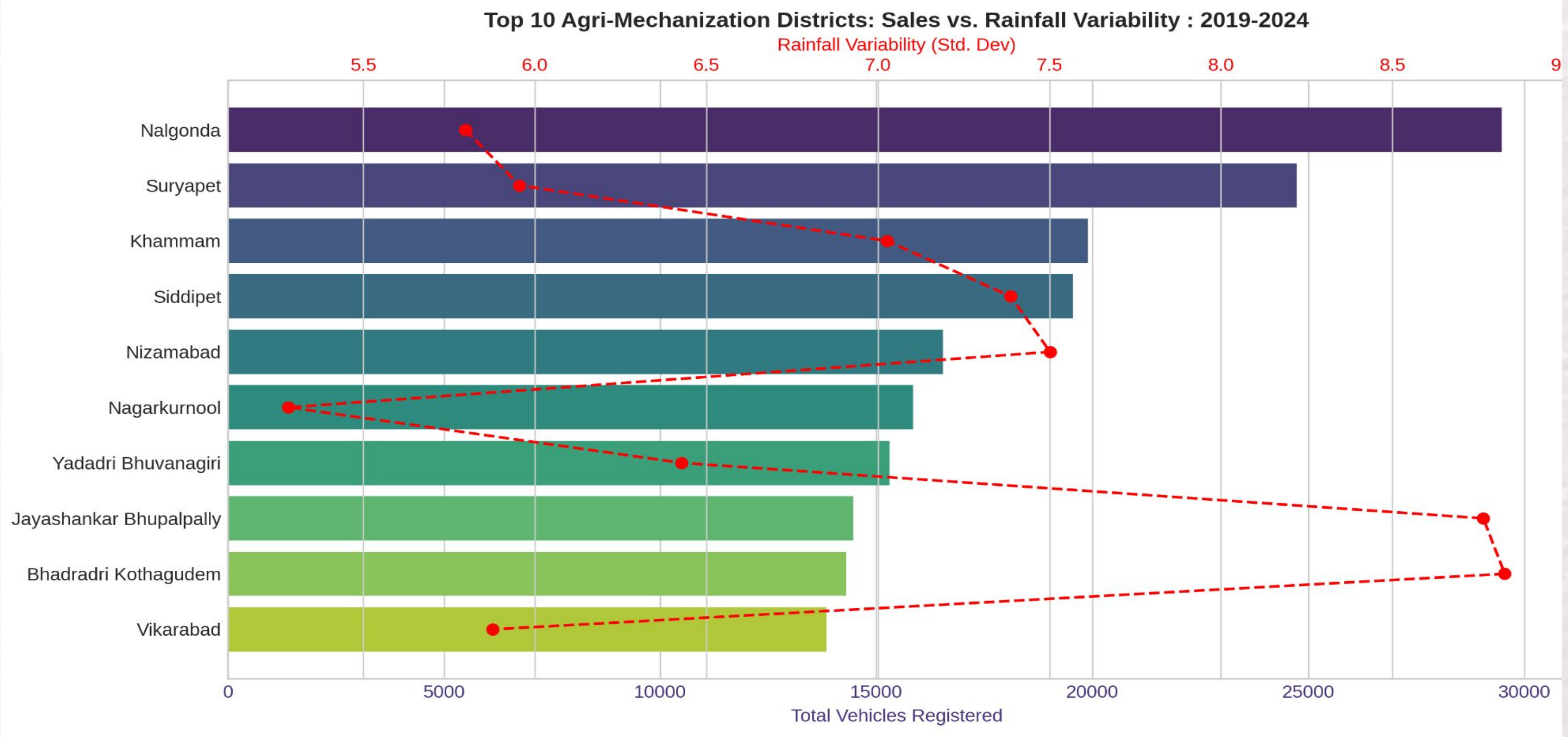


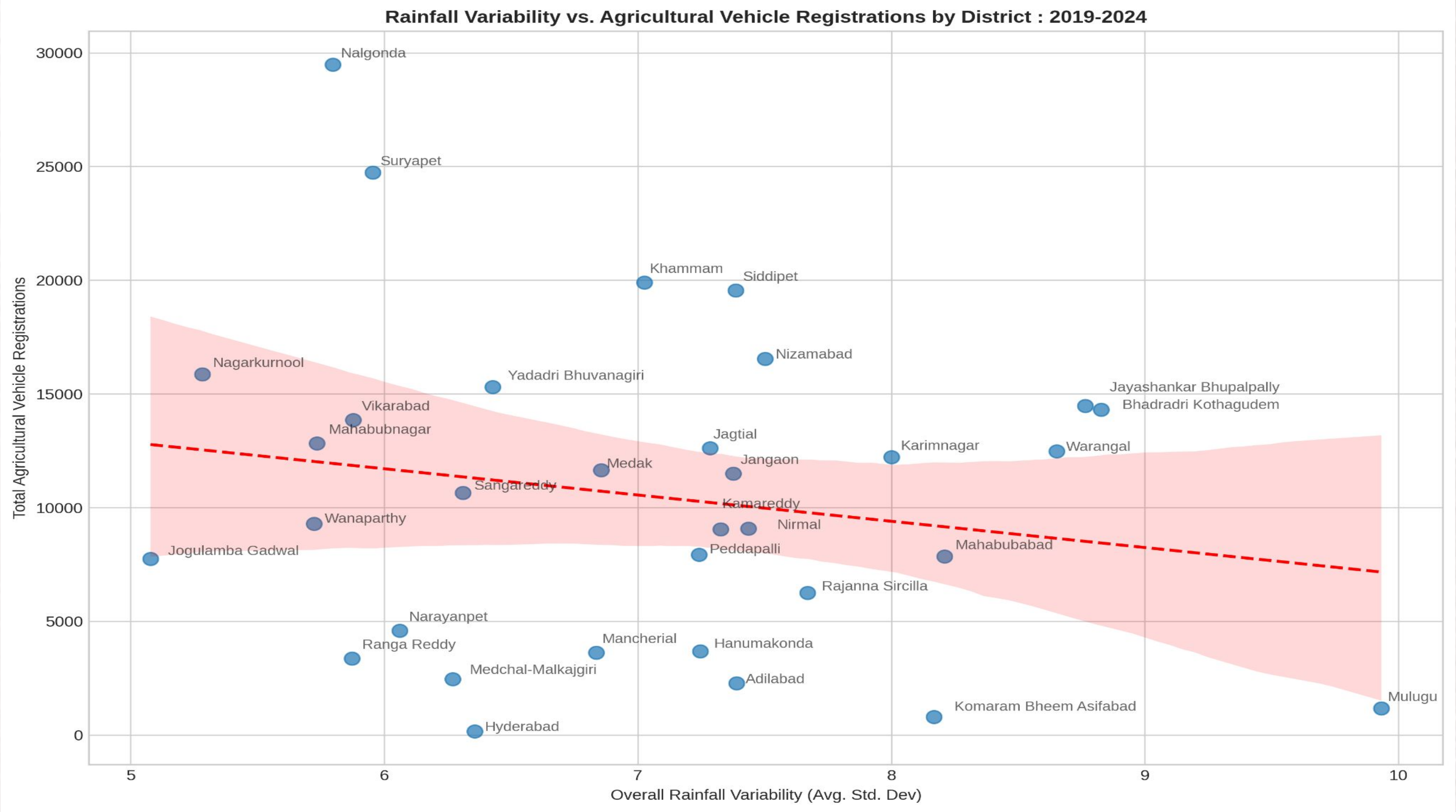


E-Stamp Digital Transformation

- The Instantaneous Transition
  - E-stamp revenue was zero until late 2020, then suddenly took over as the dominant revenue source.
- Districts generating most of the total registration revenue are major urban and peri-urban districts.
- Districts that generate most revenue and districts with most investment are highly overlapped.
- The fast e-stamp adoption shows scalable governance and digital inclusion, thus transforming document registration for the state ot fully digital.

Key Findings: Problem 3





The Agri-Mechanization Mismatch

- Districts with the highest agricultural vehicle sales shows moderate rainfall variability but not the highest.
- Districts having rain variability range 5-7 shows the most vehicle sales, and are not urban districts.
- Districts like Nalgonda and Suryapet were in the top 15 districts attracted highest investments.
- Dual-axis plot shows, districts with the most agri-vehicle sales have moderate rainfall unpredictability, not the highest.
- Scatter plot confirms, regions with the most unpredictable rainfall often have lower agri-vehicle sales.

Recommendations for Strategic Action

- **For Industrial Growth**
  - Implement a focused digitalized single-window clearance system for large scale time consuming sectors like Real Estate, Infrastructure and Textiles.
  - Address the tangible needs of MSMEs and improvised mentorship programs like T-Hub.
- **For Digital Governance**
  - Replicate the successful e-stamp model for other services including agricultural and non-commercial land registrations to improve efficiency and transparency.
  - Enhancement of digital infrastructure to ensure robust and secure data handling and protection for long term reliability and public trust.
- **For Agrarian Resilience**
  - Prioritize climate-aligned irrigation infrastructure in high-variability districts and link agricultural subsidies to water availability.
  - Re-evaluate the subsidy programs to target clusters of farmers and not on the entire state to boost the agrarian activity.

Conclusion & Future Scope

Conclusion

- **Summary** : Our data-driven analysis reframed key problems and provided clear, actionable recommendations.
- **Impact** : This approach allows the government to make informed policy decisions that balance industrial efficiency with agrarian sustainability.

Future

- Future analysis could include data on loan approvals, crop yields, and irrigation to further refine these insights.

Resources & References

- Dataset
  - Open Data Telangana
    - TS-iPASS Data
    - RTA Registration Data
  - Google Drive
    - Weather Data
    - TS-iPASS Data
    - RTA Registration Data
- Web-scraping Script : Data Collection Notebook
- Analysis Notebook : Analysis Notebook